Servo Driver HAT

From Waveshare Wiki
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Overview

Introduction

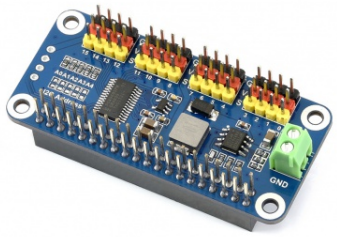
This product is a PWM/servo expansion board based on Raspberry Pi. It can expand 16-ch servo control or PWM output through the PCA9685 chip, and each channel has a 12-bit resolution. With the I2C interface for controlling, it does not require any other pins. The onboard 5V voltage regulator chip can be connected to the battery, and the maximum output current is 3A. It is suitable for controlling robotic arms and various servo robots.

Features

- Power supply: 6V-12V
- Servo voltage: 5V
- Logic voltage: 3.3V
- Driver: PCA9685
- Control interface: I2C
- Dimension: 65mm x 30mm
- Mounting hole size: 3.0mm

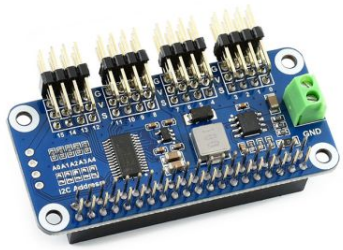
Hardware Description

Servo Driver HAT



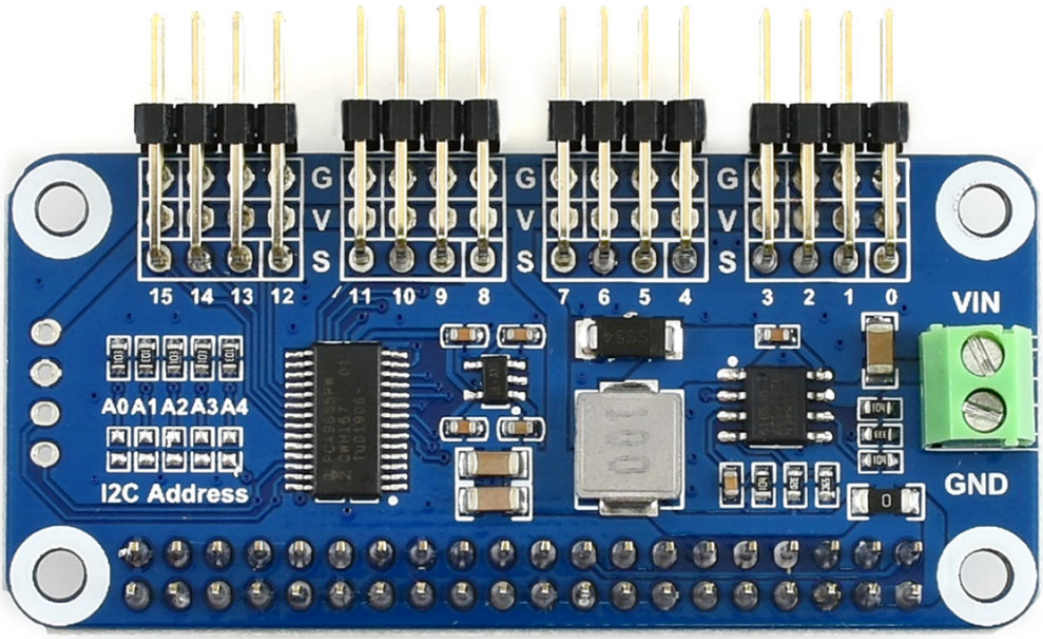
(https://www.waveshare.com/servo-driver-hat.htm)

Servo Driver HAT (B)



(https://www.waveshare.com/servo-driver-hat-b.htm)

RPI/I2C



(/wiki/File:Servo_Driver_HAT03.png)

It does not need additional power as its power supply is from Raspberry Pi.

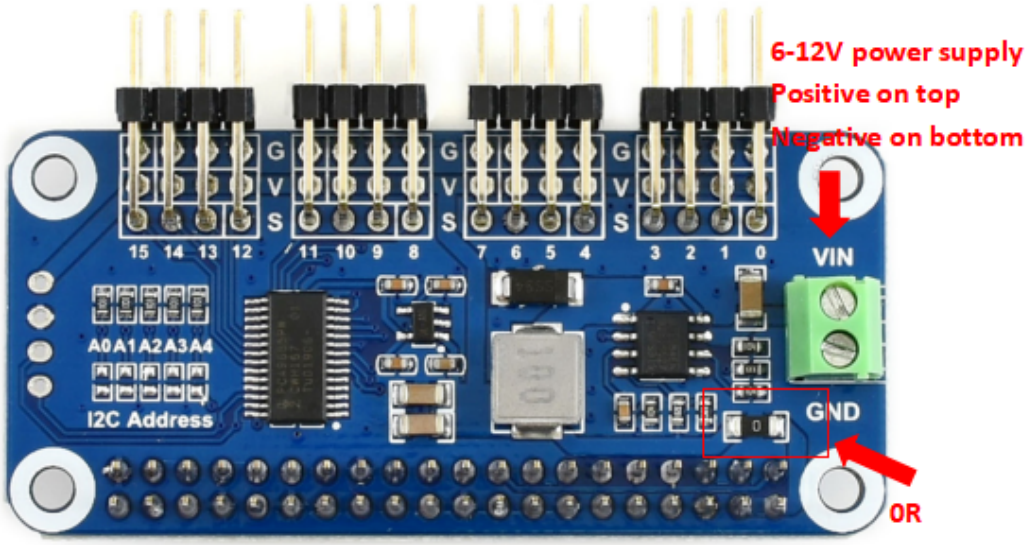
Also, you can power it by the green terminal VIN on the right side with a range of 6V-12V. When the servo and Raspberry Pi are provided by 5V from the onboard constant voltage chip, the maximum current output is 3A.

A0-A4 can be used to set the I2C device address of the PCA9685 chip and can be connected to multiple Servo Driver HAT at the same time.

The pin headers on the top are for connecting the servo. The black pin headers are for GND (mostly connected to the brown wire of the servo), the red ones are for the 5V power supply and the yellow ones are for the PWM signal cable. There are 0-15 channels and can connect to 16 servos at the same time. Please do not connect the servo cable reversely, or the servo will not rotate.

Note:

If a high-power servo is connected, the power supply may be insufficient as the whole board is powered by 5V. As both the RPI and the servo are powered by 5V, the RPI will reboot as a result of insufficient power when the power consumption is too high. In this case, we should remove the onboard 0R resistor and connect the external power by VIN (6-12V).



(/wiki/

File:Servo_Driver_HAT1.png)

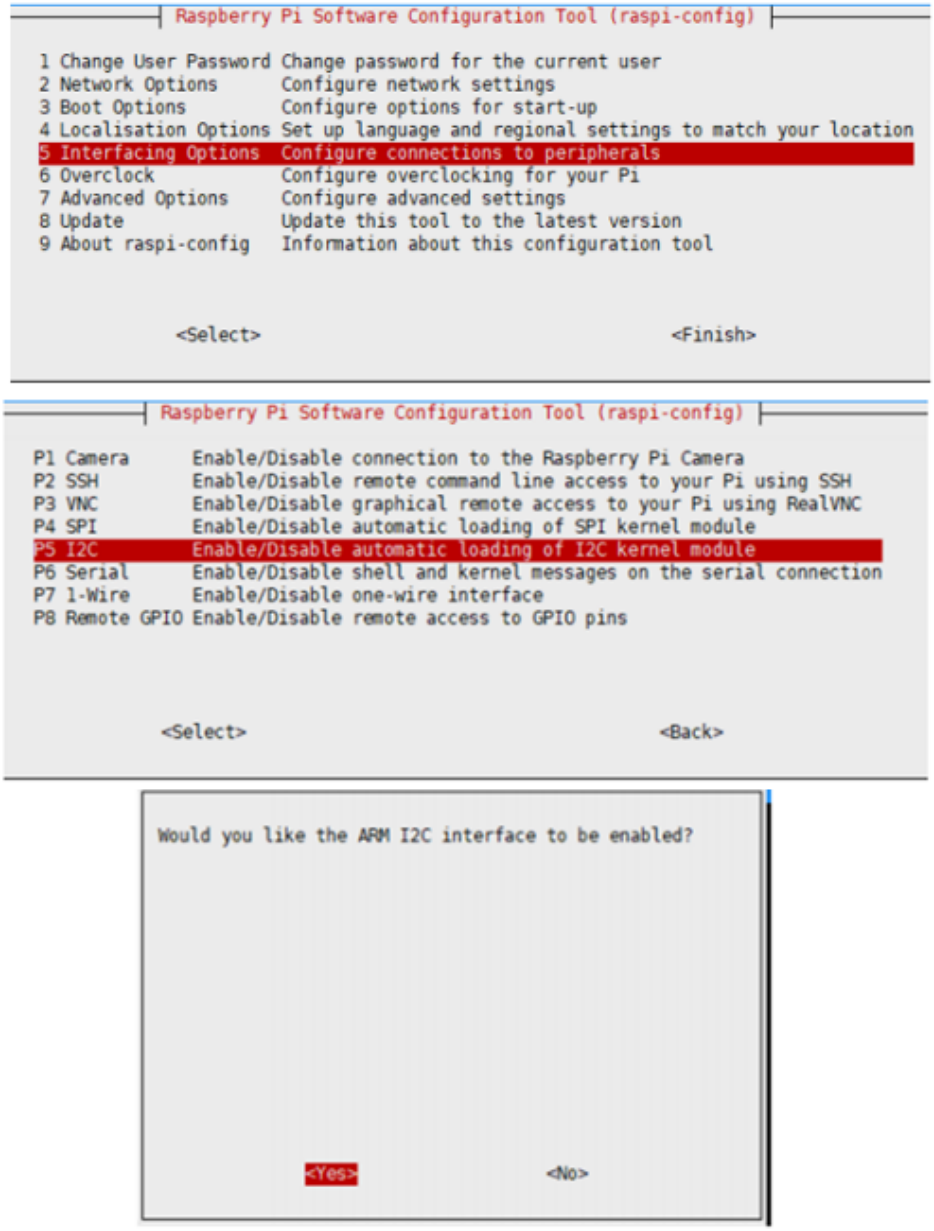
Using with Raspberry Pi

To use this module, we provide python examples for test PCA9685, WiFi remote control and bluetooth remote control.

Enable I2C Interface

- Open the Raspberry Pi terminal and input the following command to enter the configuration interface:

```
sudo raspi-config
Select Interacting Options ->I2C ->yes to start the I2C kernel driver
```



(/wiki/

File:RPI_open_i2c.png)

Then restart Raspberry Pi:

```
sudo reboot
```

Install libraries

```
sudo apt-get update
sudo apt-get install python-pip
sudo pip install RPi.GPIO
sudo apt-get install python-smbus
```

Downalod the demo codes and unzip

You should start the Raspberry Pi, open a terminal and run the following commands:

```
sudo apt-get install p7zip-full
wget https://www.waveshare.com/w/upload/6/6c/Servo_Driver_HAT.7z
7zr x Servo_Driver_HAT.7z -r -o./Servo_Driver_HAT
sudo chmod 777 -R Servo_Driver_HAT
cd Servo_Driver_HAT/Raspberry\ Pi/
```

Python Examples

Open a terminal and runt the following comamnds:

```
#For python2
cd ~/Servo_Driver_HAT/Raspberry\ Pi/
cd python/
sudo python PCA9685.py
#For python3
cd ~/Servo_Driver_HAT/Raspberry\ Pi/
cd python3/
sudo python3 PCA9685.py
```

Expected result: Connect a servo to Channel 0, the servo will rotate.

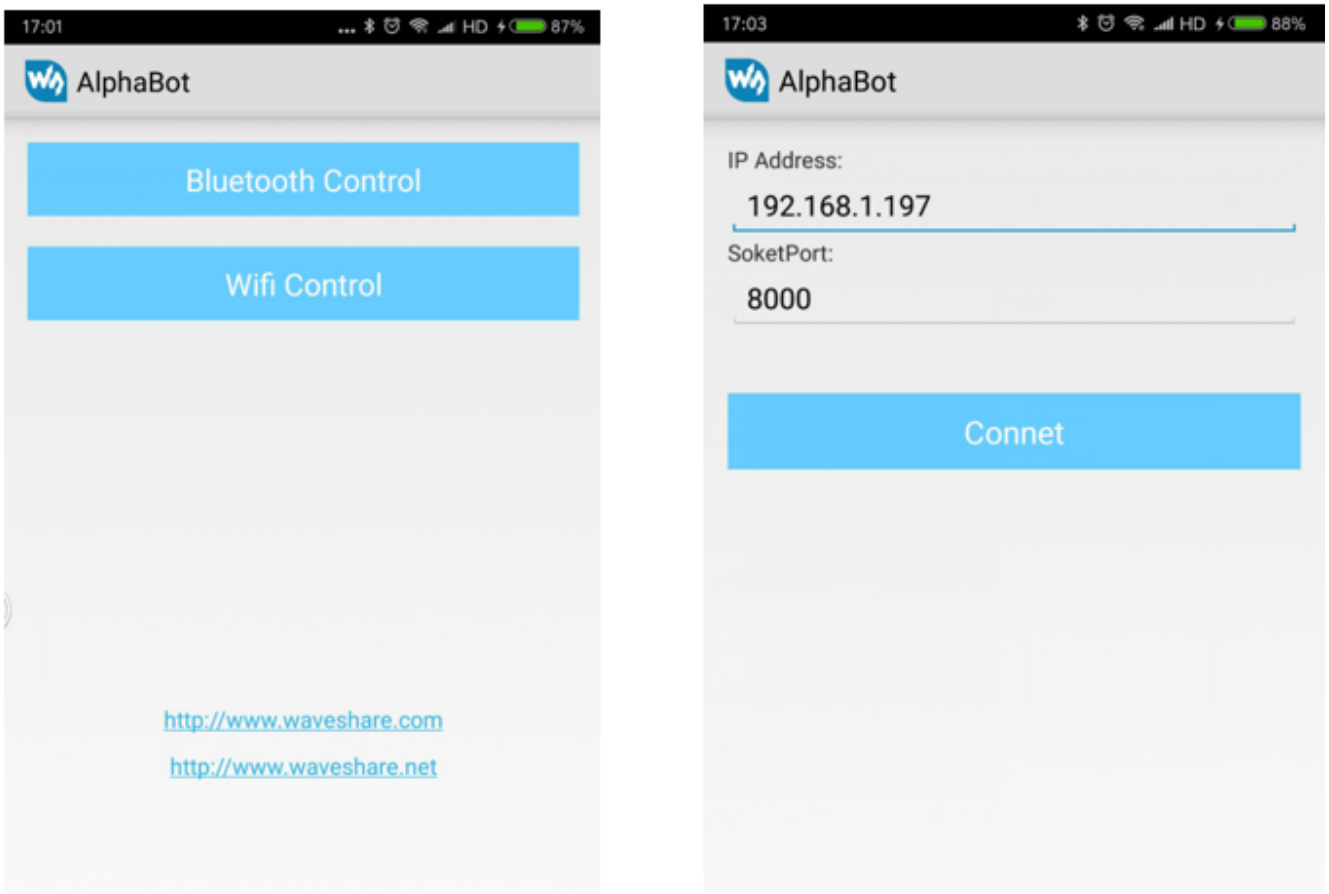
WIFI Remote Control

Open a terminal and run the following commands:

```
cd Wifi-Control/
sudo python main.py
```

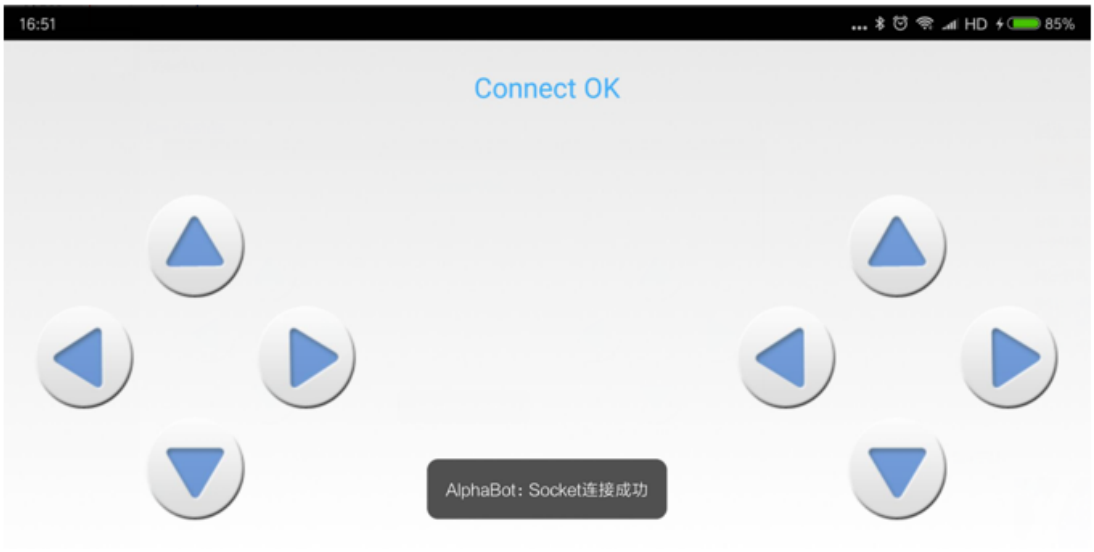
Please connect the Raspberry Pi and telephone to the same WLAN network. Data will be transmitted by TCP protocol. After running the program, the IP address will be displayed and the port number is 8000.

Open the APP and choose the WIFI control, input the IP address and the port then connect.



(/wiki/File:Servo_driver_hat_wifi.png)

It enters the control page if connected successfully. You can click the button to control the four servos (Channel 0 to Channel 4).

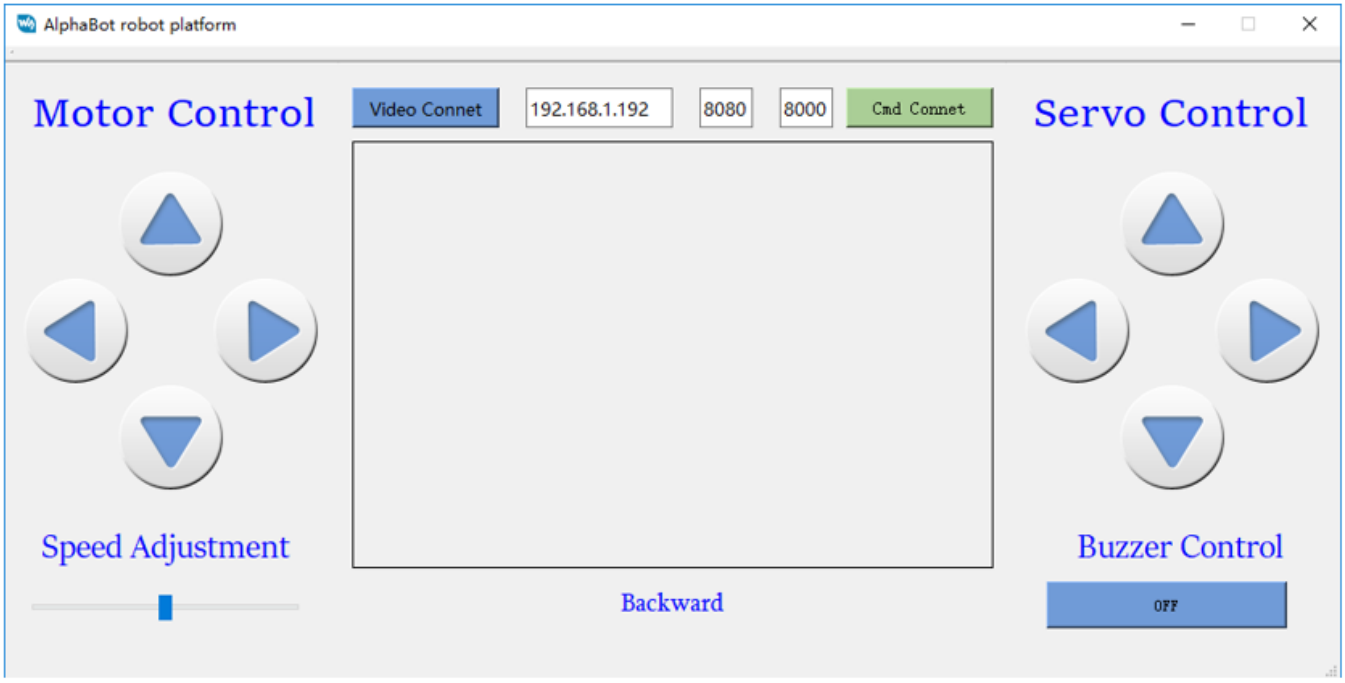



```
pi@raspberrypi:~/Servo_Driver_HAT/python/Wifi-Control $ sudo python main.py
start
192.168.1.192

server is running....
('got connection from ', ('192.168.1.184', 59089))
Forward
Stop
Backward
Stop
```

(/wiki/File:Servo_driver_hat_wifi1.png)

You can also connect it with QT software on Windows PC.



(/wiki/File:Servo_driver_hat_wifi2.png)

Note: You can download the APP and the QT software from the Resources part.

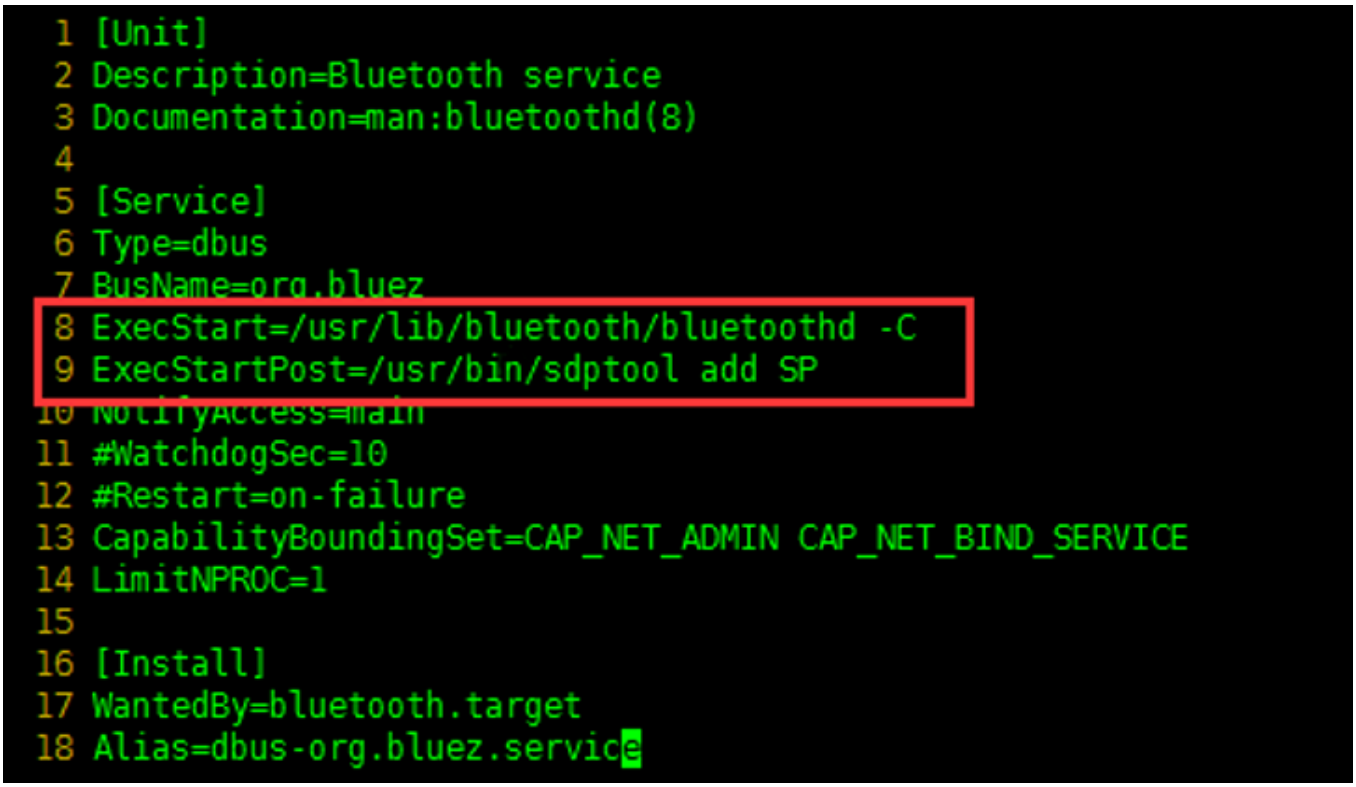
Bluetooth Remote Program

Execute

```
sudo apt-get update
sudo apt-get upgrade -y
sudo apt-get dist-upgrade -y
sudo apt-get install pi-bluetooth bluez bluez-firmware blueman
sudo usermod -G bluetooth -a pi
sudo vi /etc/systemd/system/dbus-org.bluez.service
sudo reboot
```

Start/add SPP, and turn on the Bluetooth device.

```
sudo vi /etc/systemd/system/dbus-org.bluez.service
```



(/wiki/File:Servo_driver_hat_bt.png)

After restarting the Raspberry Pi, enter the hciconfig command (similar to the ifconfig command) to check the Bluetooth service.

```
pi@raspberrypi:~ $ sudo hciconfig
hci0:   Type: BR/EDR  Bus: UART
        BD Address: B8:27:EB:2D:00:87  ACL MTU: 1021:8  SCO MTU: 64:1
        UP RUNNING
        RX bytes:717 acl:0 sco:0 events:42 errors:0
        TX bytes:1532 acl:0 sco:0 commands:42 errors:0
```

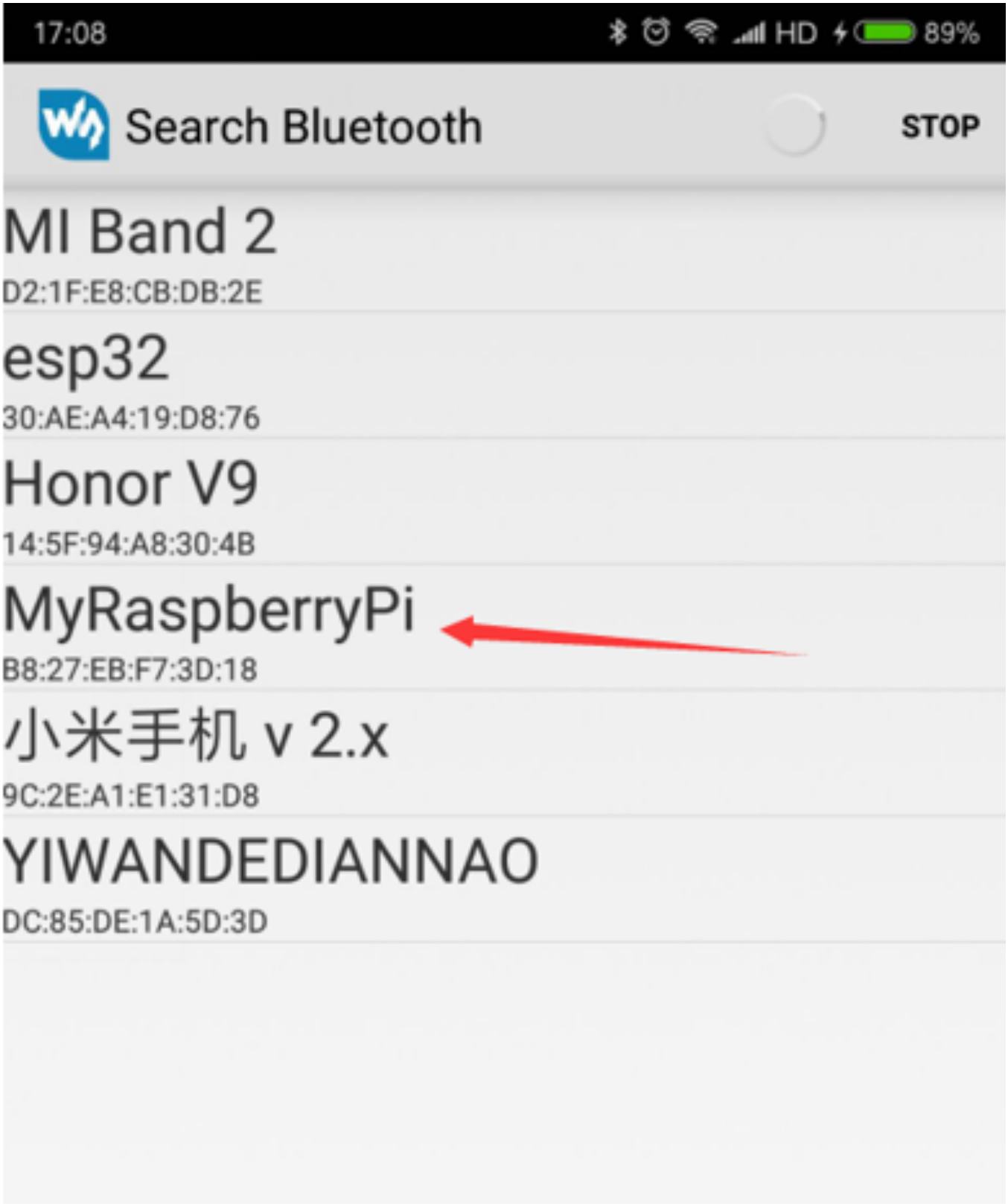
(/wiki/File:Servo_driver_hat_bt2.png)

If you see the hci0 device, the Bluetooth has been turned on. If not then no Bluetooth device is recognized.

Run the following command to start the program.

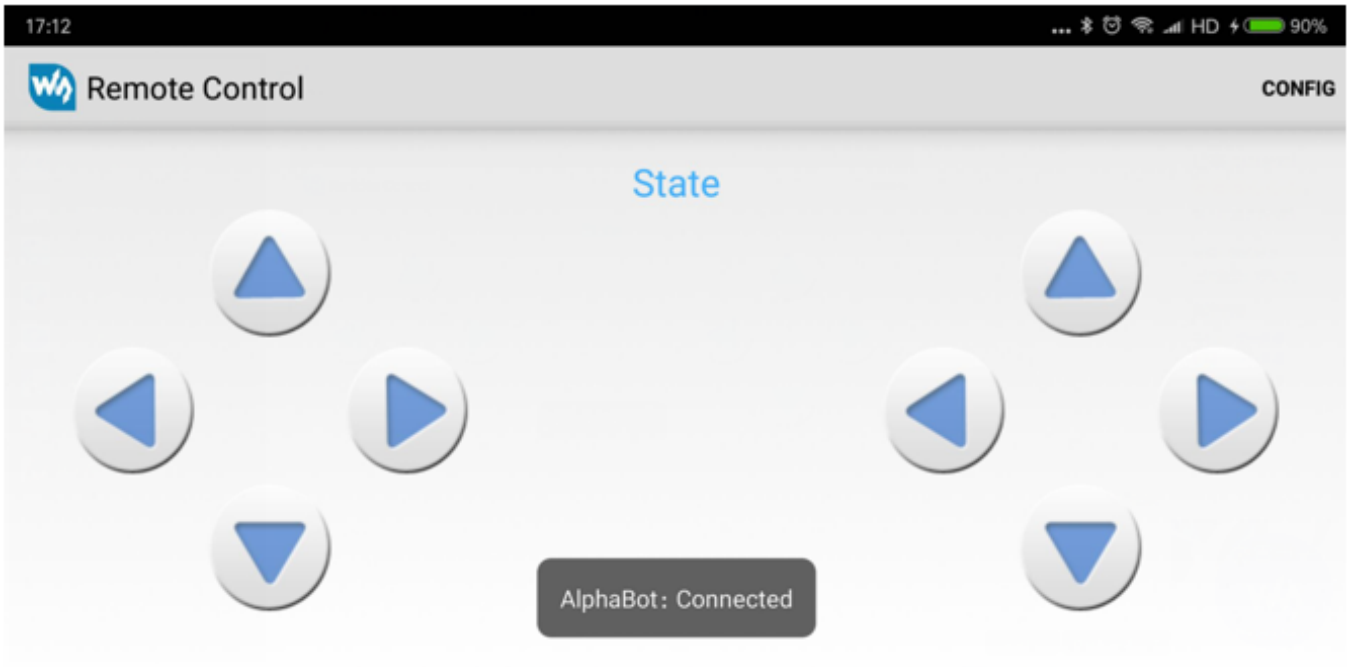
```
cd Servo_Driver_HAT/python/Bluetooth-Control
sudo ./Bluetooth.sh
```

After the program runs, you will be prompted to wait for the Bluetooth connection, choose the Bluetooth remote control on the mobile app, click scan, discovers the Raspberry Pi device, and connects.



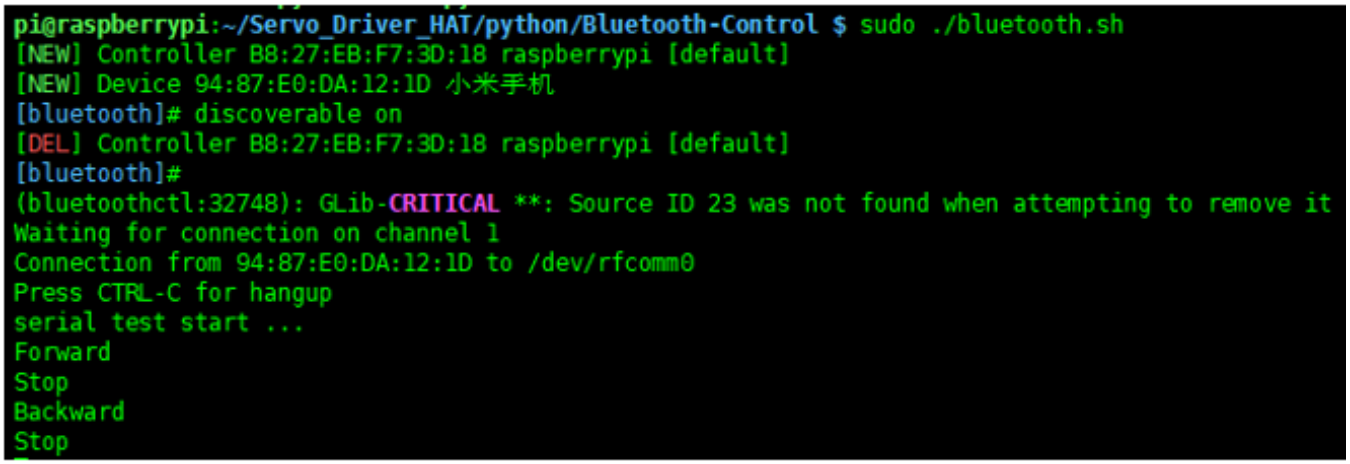
(/wiki/File:Servo_driver_hat_bt3.png)

After connecting the program, enter the control interface, and click the button to remotely control the forward and reverse rotation of the servos of channels 0~4.



(/wiki/File:Servo_driver_hat_bt4.png)

Note: If the received command is incorrect, you may need to configure the command sent when the corresponding button is pressed and released in the APP.

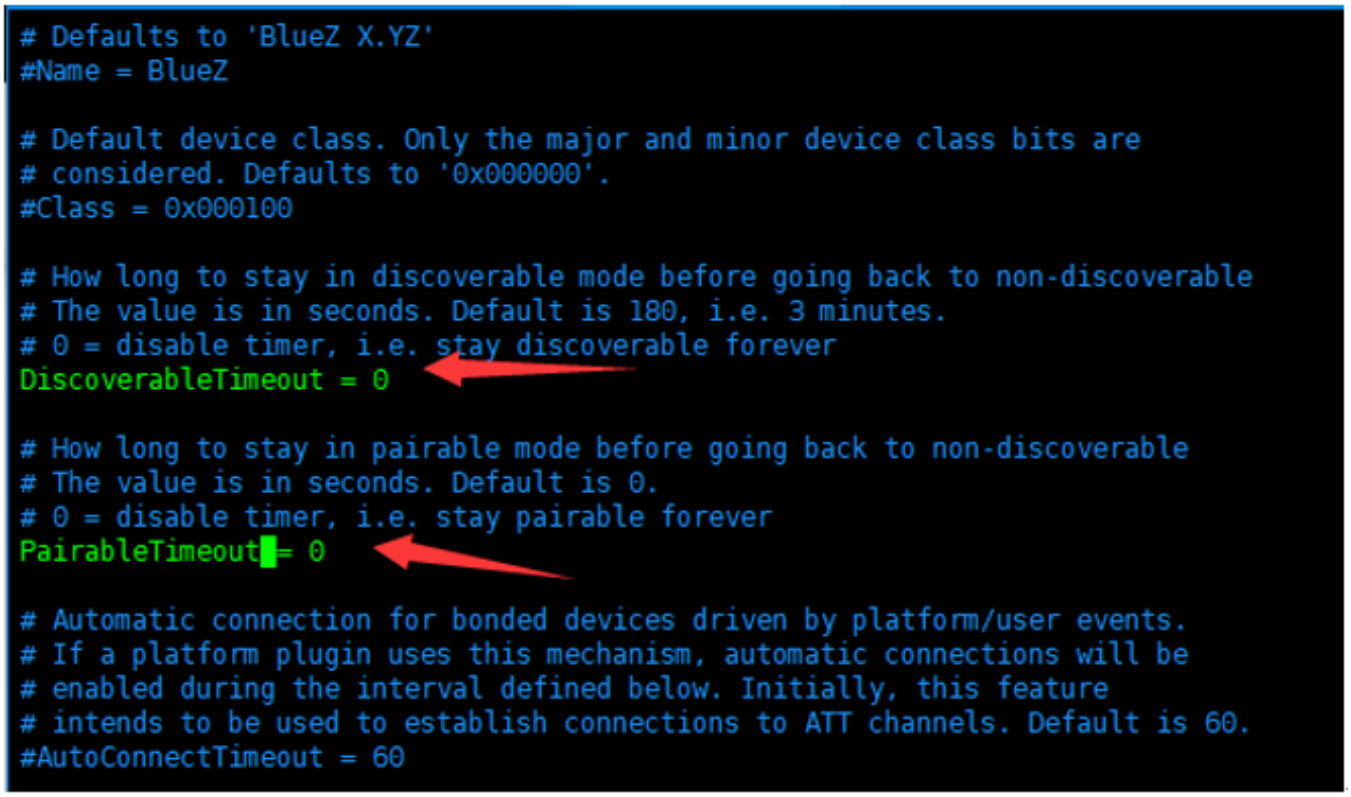


(/wiki/File:Servo_driver_hat_bt5.png)

Note: The default discovery time of Raspberry Pi is only 180s. If you want to always discover and pair, you can configure it by modifying the following files.

```
sudo vi /etc/bluetooth/main.conf
```

Find the following two statements and uncomment them.



(/wiki/File:Servo_driver_hat_bt6.png)

The program is compatible with python3, just replace python with python3 when running.

For more information on Raspberry Pi wifi and Bluetooth remote control, please refer to AlphaBot2:

http://www.waveshare.com/wiki/AlphaBot2 (http://www.waveshare.com/wiki/AlphaBot2)

https://www.waveshare.com/wiki/AlphaBot2 (https://www.waveshare.com/wiki/AlphaBot2)

Using with Jetson Nano

Install Library

- Enable the terminal interface and enter the following command to install the library.

```
sudo apt-get update
sudo apt-get install python3-pip
sudo pip3 install Jetson.GPIO
sudo groupadd -f -r gpio
sudo usermod -a -G gpio your_user_name
sudo cp /opt/nvidia/jetson-gpio/etc/99-gpio.rules /etc/udev/rules.d/
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Note your_user_name is the name you used such as waveshare.

- Install I2C

```
sudo apt-get install python-smbus
```

- Install image processing library:

```
sudo apt-get install python3-pil
sudo apt-get install python3-numpy
```

Downlaod the example program & unzip to the specified directory

```
sudo apt-get install p7zip
wget https://www.waveshare.com/w/upload/6/6c/Servo_Driver_HAT.7z
7zr x Servo_Driver_HAT.7z -r -o./Servo_Driver_HAT
sudo chmod 777 -R Servo_Driver_HAT
cd Servo_Driver_HAT/Jetson\ Nano/
```

Run the test demo

- python 2

```
cd python2/
sudo python test.py
```

- python 3

```
cd python3/
sudo python3 test.py
```

Working with VisionFive2

Installing the Corresponding Library File

```
apt-get install pip
pip install VisionFive.gpio
apt-get install python3-smbus
```

Demo Download

```
apt-get install p7zip-full
wget https://files.waveshare.com/upload/6/6c/Servo_Driver_HAT.7z
7zr x Servo_Driver_HAT.7z -r -o./Servo_Driver_HAT
cd Servo_Driver_HAT/VisionFive2/python
```

Direct Use

```
python3 PCA9685.py
```

Expected effect: The servo of channel 0 turns from 0 degrees to 180 degrees, and then turns from 180 degrees to 0 degrees again, repeating the cycle continuously.

Wifi Remote Control

```
cd Wifi-Control/
python3 main.py
```

For details, you can refer to #WIFI Remote Control.

Note:

1. Control can only be done within the same LAN (Local Area Network).
2. There is a feature (a user-friendly bug) that allows the device to be used with only an Ethernet connection when there is no WiFi module connected. However, after connecting a WiFi module and running the demo once, if you unplug the WiFi module and run the demo again, an error may occur. Simply restarting the device should resolve this issue.

Resources

Document

- User Manual (https://files.waveshare.com/upload/1/1b/Servo_Driver_HAT_User_Manual_EN.pdf)
- Schematic (https://files.waveshare.com/upload/a/a2/Servo_Driver_HAT_Schematic_.pdf)
- Demo code (https://www.waveshare.com/wiki/File:Servo_Driver_HAT.7z)
- Servo Driver HAT 3D Drawing (https://files.waveshare.com/wiki/Servo%20Driver%20HAT/3D-Drawings/Servo-Driver-HAT.zip)
- Servo Driver HAT (B) 3D Drawing (https://files.waveshare.com/wiki/Servo-Driver-HAT-(B)/3D-Drawings/Servo-Driver-HAT-(B).zip)

Software

- QT software for PC (https://files.waveshare.com/upload/a/a9/AlphaBot_Qt.7z)
- App for Android (https://files.waveshare.com/upload/5/56/AlphaBot_LITE.rar)
- App source code (https://files.waveshare.com/upload/3/39/AlphaBot_LITE_Code.7z)

Datasheet

- PCA96 (https://files.waveshare.com/upload/6/68/PCA96_datasheet.pdf)
- MP1584 (https://files.waveshare.com/upload/d/d3/MP1584.pdf)
- RT9193 (https://files.waveshare.com/upload/f/f6/RT9193.pdf)

Third-Parties Porject

- Node-Red project shared by liot2K (https://github.com/iiot2k/node-red-contrib-iiot-rpi-pca9685)

FAQ

Question:Can the angle of rotation be precisely controlled?

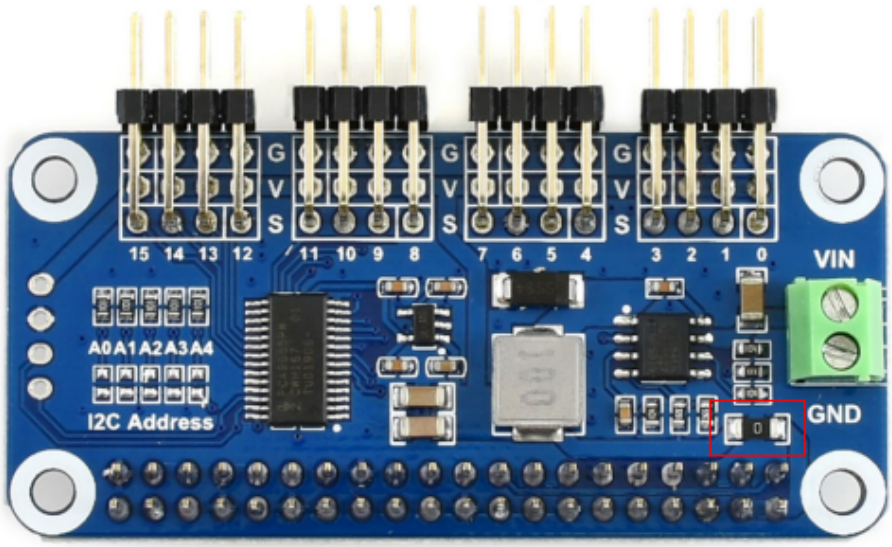
Answer:
No, this is just entry-level.

Question:Why does it shake?

Answer:
It is normal for it to shake slightly. Since the actual angle of the servo is smaller than the minimum physical angle, there will be a current to maintain its angle, which will cause shaking.

Question:Why does the Raspberry Pi restart when controlling high-power servos?

Answer:
Since the Raspberry Pi provides 5V to the module by default, if the control servo power is too large (such as MG996R, DS3120MG), the 5V of the Raspberry Pi will be pulled down. You can remove the onboard 0R resistor and provide 6-12V for the VIN.



(/wiki/

File:Servo_Driver_HAT06.png)

Question:Can it be used to control a stepper motor?

Answer:
No, this output is PWM and cannot control the servo.

Question:Servo squeaking and accompanying heat?

Answer:
The SG90 is a servo for beginners. As the potentiometer inside is inaccurate, it will loosen after use for a period, which causes the gear to be inconsistent with the theoretical rotation angle, resulting in noise and heat.

Question:Why do two I2C device addresses appear when scanning an I2C address?

Answer:
The control chip used corresponds to PCA9685. When powered on, there are two I2C addresses, one is the address configured according to the onboard resistance, the default is 0X40, and the other is 0X70, which is configured by the ALLCALLADR register. You can run the demo again to check the value of the register.

Question:Can two identical Servo Driver HATs be stacked?

Answer:
It can be stacked, but the resistance of the I2C address needs to be changed. In the I2C Address on the left, solder the default upper resistance to the bottom with a soldering iron. Different combinations generate different I2C address combinations, and there are a total of 32 combinations of 2 to the 5th power.

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue. Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

Submit Now (https://service.waveshare.com/)

Retrieved from "https://www.waveshare.com/w/index.php?title=Servo_Driver_HAT&oldid=79075 (https://www.waveshare.com/w/index.php?title=Servo_Driver_HAT&oldid=79075)"